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COVER STORY

Teen builds an award-winning virtual reality prototype thanks to free MIT courses

When Freesia Gaul discovered MIT Open Learning's OpenCourseWare at just 14 years old, it opened up a world of learning far beyond what her classrooms could offer.

Stefanie Koperniak | MIT Open Learning · MIT News Education

Her parents had started a skiing company, and the seasonal work meant that Gaul had to change schools every six months. Growing up in small towns in Australia and Canada, she relied on the internet to fuel her curiosity.

"I went to 13 different schools, which was hard because you're in a different educational system every single time," says Gaul. "That's one of the reasons I gravitated toward online learning and teaching myself. Knowledge is something that exists beyond a curriculum."

The small towns she lived in often didn't have a lot of resources, she says, so a computer served as a main tool for learning. She enjoyed engaging with Wikipedia, ultimately researching topics and writing and editing content for pages. In 2018, she discovered MIT OpenCourseWare, part of MIT Open Learning, and took her first course. OpenCourseWare offers free, online, open educational resources from more than 2,500 MIT undergraduate and graduate courses.

"I really got started with the OpenCourseWare introductory electrical engineering classes, because I couldn't find anything else quite like it online," says Gaul, who was initially drawn to courses on circuits and electronics, such as 6.002 (Circuits and Electronics) and 6.01SC (Introduction to Electrical Engineering and Computer Science). "It really helped me in terms of understanding how electrical engineering worked in a practical sense, and I just started modding things."

In true MIT "mens et manus" ("mind and hand") fashion, Gaul spent much of her childhood building and inventing, especially when she was able to access a 3D printer. She says that a highlight was when she built a life-sized, working version of a Mario Kart, constructed out of materials she had printed.

Gaul calls herself a "serial learner," and has taken many OpenCourseWare courses. In addition to classes on circuits and electronics, she also took courses in linear algebra, calculus, and quantum physics — in which she took a particular interest.

When she was 15, she participated in Qubit by Qubit. Hosted by The Coding School, in collaboration with universities (including MIT) and tech companies, this two-semester course introduces high schoolers to quantum computing and quantum physics.

During that time she started a blog called On Zero, representing the "zero state" of a qubit. "The 'zero state' in a quantum computer is the representation of creativity from nothing, infinite possibilities," says Gaul. For the blog, she found different topics and researched them in depth. She would think of a topic or question, such as "What is color?" and then explore it in great detail. What she learned eventually led her to start asking questions such as "What is a hamiltonian?" and teaching quantum physics alongside PhDs.

Building on these interests, Gaul chose to study quantum engineering at the University of New South Wales. She notes that on her first day of university, she participated in iQuHack, the MIT Quantum Hackathon. Her team worked to find a new way to approximate the value of a hyperbolic function using quantum logic, and received an honorable mention for "exceptional creativity."

dir="ltr">Gaul's passion for making things continued during her college days, especially in terms of innovating to solve a problem. When she found herself on a train, wanting to code a personal website on a computer with a dying battery, she wondered if there might be a way to make a glove that can act as a type of Bluetooth keyboard — essentially creating a way to type in the air. In her spare time, she started working on such a device, ultimately finding a less expensive way to build a lightweight, haptic, gesture-tracking glove with applications for virtual reality (VR) and robotics.

dir="ltr">Gaul says she has always had an interest in VR, using it to create her own worlds, reconstruct an old childhood house, and play Dungeons and Dragons with friends. She discovered a way to put into a glove some small linear resonant actuators, which can be found in a smartphone or gaming controller, and map to any object in VR so that the user can feel it.

dir="ltr">An early prototype that Gaul put together in her dorm room received a lot of attention on YouTube. She went on to win the People's Choice award for it at the SxSW Sydney 2025 Tech and Innovation Festival. This design also sparked her co-founding of the tech startup On Zero , named after her childhood blog dedicated to the love of creation from nothing.

dir="ltr">Gaul sees the device, in general, as a way of “paying it forward,” making improved human-computer interaction available to many — from young students to professional technologists. She hopes to enable creative freedom in as many as she can. “The mind is just such a fun thing. I want to empower others to have the freedom to follow their curiosity, even if it's pointless on paper.

dir="ltr">“I’ve benefited from people going far beyond what they needed to do to help me,” says Gaul. “I see OpenCourseWare as a part of that. The free courses gave me a solid foundation of knowledge and problem-solving abilities. Without these, it wouldn’t be possible to do what I’m doing now.”

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FEATURES

Roots of Righteousness: A Glimpse into Ethics as the Moral Fabric of India

George Levrier-Jones · History in 28 Minutes

Ethics are moral principles that govern a person's behavior in a society. In very simple terms, they constitute the rights and wrongs that guide people's conduct. The Indian ethics are generally connected to the principle of *anekantavada* or many-sidedness, emphasizing that there is "no absolute truth" and no neatly defined binaries of right and wrong. They are based on factors such as the person practicing them, the situation, and the time at which they are practised. Indian ethics focus on accepting and encouraging diverse thoughts and beliefs, hence propagating "unity in diversity" and "diversity in unity."

"It is surely extremely strange that whenever, either in Greek, or in Chinese, or in Persian, or in Arab writings, we meet with any attempts at describing the distinguishing features in the national character of the Indians, regard for truth and justice should always be mentioned first." — F. Max Müller, Sanskritist and philologist (1882)

Composed between 1500–1000 BCE, the Rig Veda mentions the concept of *Ritam* (cosmic order) through which the physical and the social worlds are sustained. *Ritam* can be understood as the sense of righteousness. It later developed into the concept of *Satyam* (truth), with strong ethical implications. *Dharma*, the building block of Indian ethics, has been translated from the Rig Veda to refer to words such as justice, duty, righteousness, and order, among others. It is important to note that *dharma* is a multifaceted concept and does not denote a single idea or meaning.

Composed between 800 and 500 BCE, the Upanishads reveal further developments in Indian ethical thought. For example, *Aham Brahma asmi* (1.4.10; *Brihadaranyaka Upanishad*) translates to "I am Brahman (the Absolute)" and can be understood as: A person is a part of God (and not separate from this universal consciousness). It is essential to grow cognizant of this identity. Further, the Upanishads highlight that every person has a distinct nature (*svabhava*), function, truth, and path (*svadharma*), echoing the concept of *anekantavada*. The Puranas also propagate notions like all creation is interconnected and that one can be happy when all are happy.

The Ramayana and Mahabharata stories underline the idea that ethics are complex. Several dilemmas shape the events of these epics and put forth the question: How to decide what is *dharma* in different situations? The central message of the Bhagavad Gita, part of the Mahabharata, is *Nishkama Karma*, meaning desireless action. It propagates the concept that people should work without any expectation of results. What matters ultimately is the inner growth of the individual, which can be achieved when one's actions align with one's *dharma*. *Nishkama Karma* forms the basis of *Karma Yoga*, a spiritual practice of selfless action that can lead a person to liberation (*moksha*).

Lord Krishna emphasized that *dharma* cannot be practiced in passivity; several situations arise when a person needs to take a particular side in life. Some examples showcasing the sophisticated nature of *dharma* include the following:

Dronacharya : In the Mahabharata, Drona was a master of advanced military arts, including divine weapons. The Pandavas decided to use Drona's only weakness, his son Ashwatthama, to save many lives in the war. The Pandava Bhima killed an elephant named Ashwatthama and spread the news of his death. Drona asked the Pandava Yudhishtira (known for always speaking the truth), who clearly said that Ashwatthama was killed. It is said that Krishna suppressed the word 'elephant' by blowing the conch. The news of the death of his son eventually led to the slaying of Drona. This story emphasizes that if an untruth saves lives, then perhaps ethics is not about always telling the truth.

Vibhishana and Kumbhakarna : Ravana's brothers Vibhishana and Kumbhakarna in the Ramayana skillfully demonstrated that there is no single path to *dharma* and no single way of solving an ethical dilemma. Kumbhakarna adhered to the *dharma* of loyalty to his kin. At the same time, Vibhishana chose to follow the *dharma* of saving the people from the evil of his brother Ravana by opposing his kin and supporting Rama.

Buddhism originated in the Indian subcontinent and mentions a noble eightfold path encompassing the 'right' vision, intention/aspiration, speech, action, livelihood, effort, mindfulness, and concentration. Once Buddha preached, "... whenever someone abuses us, we can either choose to accept or decline that anger. Our response will decide who owns and keeps the bad and negative feelings." This story beautifully encourages the practice of the right speech, action, and mindfulness. Buddhism also talks about having compassion for others. Aligning with this idea, Buddhist economics studies the production, distribution, and consumption of goods/services by changing the focus from self-interest to no-self (generosity), with 'right' livelihood and sustainability. While traditional economics emphasizes maximizing profits, Buddhist economics aims to minimize suffering (losses) for all.

Jainism also originated in the Indian subcontinent and mentions the *Triratna* (three jewels) as the 'right' faith, knowledge, and walk. *Ahimsa* (non-violence), *Satya* (truth), *asteya* (non-stealing), *brahmacharya* (chastity), and *aparigraha* (Non-possession) are its five ethical codes. Detachment (non-attachment or non-possession) is one of its main morals. It can serve as the means to attain the realization of one's self. Jainism's concept of violence, acceptable only when absolutely necessary for self-defense, appears to echo with Lord Krishna's advice from the Mahabharata that *dharma* cannot be practiced in passivity (at least, in certain situations according to Jainism).

Composed around the 5th century CE by Thiruvalluvar and consisting of 1,330 short couplets, the *Kural* is a classic Tamil language text. It is considered a great work

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Code reviews for fun and profit

Monica Dinculescu

id="stats-a-preamble">Stats: a preamble

I've been reading too much about March Madness brackets, so I thought I had to run some numbers around here like the cool kids do. Get your umbrella out, it's about to rain cold facts.

In the history of time, Chromium has had 205,095 commits made by 1,943 contributors representing 7,431,088 lines of code. In the last 30 days, there have been 5021 commits, by 637 contributors, including 53 new hoomans.

I did some advanced Nate Silver analysis here for you, and that's at least 167 commits and 1+ new committers a day. On average, that's at least 7 commits an hour. Every hour. All of the hours.

That's an imperial ton of new code being added, by what it seems like new people. Imagine if everyone could commit code willy-nilly. Are you imagining a minefield? You should.

id="code-reviews-ftw">Code reviews ftw

Good news for our browser using audience! Chromium isn't a minefield, and on top of it, has pretty awesome looking code. This comes from the fact that any code changes need to be reviewed and blessed before they can land on the master branch. More eyes means less bugs means you're less likely to commit broken code and break the internet. And you really don't want to break the internet.

Even if you have tests, and everything is going your way, you can write correct, but genuinely shitty code. 7 million lines of kinda-shitty code is not something anyone wants to work with, and are worth investing a little time in fixing.

Code reviews also bring up the bus factor, which is my favourite sinister nerd metaphor. You know, the buuuuus factor. The number of people that can get run over by a bus on a team before that team is royally and epically screwed. If all the code that you write has been closely read by a different person, then you're probably ok getting run over by a bus every once in a while. But still, you probably shouldn't. Who would feed your cat?

id="consistent-code-is-the-best-code">Consistent code is the best code

Code style guides are sooper neat, and are a huge part of code reviews, because ain't nobody got time to argue about braces. You can spend that time arguing about imperative vs. functional languages, which is a much better use of everyone's time. Having a strict and detailed style guide means that, even though it's probably going to piss off some very opinionated people, all of the code will look the same, all the time, regardless of who wrote it. You can jump into any area of Chromium and feel at home, because nobody went crazy with the whitespace.

And trust me, it only takes a week to get over the wrong kind of braces.

id="dont-be-scared">Don't be scared

We've reached the point in the blog post where I confess I am a terrible code reviewer. I am incredibly scared of reviewing code I haven't written. Guess what: in the case of Chromium, that's most of it.

Reviewers are usually picked from a list of owners, which is a group of people that is intimate enough with that area of code that they've taken it out to dinner a couple of times. They're the ones that have the final say on whether the code is ok, and who make sure that little neighbourhood of code isn't a minefield. Even if you're not an owner, your team mates will probably ask you for a first review of their code, to make the owners' lives easier. You will be tempted to panic and not want to take responsibility for it. You will be tempted to run away from confrontation and agree with all of their changes.

Don't. It doesn't help anyone, and it will be like you're not even there. You were asked for a review because people trust your judgement and value your opinion. So give it. The worst thing that can happen is that they will disagree, and you will have a polite conversation about it. The best thing that can happen is amazingly awesome code. So, have a little courage, and be the little reviewer that could.

id="dont-be-a-jerk">Don't be a jerk

Jerks are the worst code reviewers. Generally, people tend to get very defensive when faced with criticism, and they'll get exponentially more defensive if that criticism comes in a harsh, patronizing voice. Defensive people aren't open to discussions, and it will make the review experience painful for everyone.

Don't be a control freak either. You might disagree with the names of variables and functions, but unless you have good suggestions, you might want to consider conceding those points.

The moral here is: the code you're reviewing was written by a smart human. Treat them like one.

id="what-makes-a-great-code-reviewer">What makes a great code reviewer

Good code reviewers are diligent: they enforce the style guide, they make sure you've documented the new code, and they aren't scared of making you shave a yak or four if that's needed (this includes both yaks that you have conjured and the ones you've accidentally stumbled upon).

Even better reviewers will try to help the author of the code learn. This is a very hard stack of plates to balance: on one hand you don't want to be lazy and offer really vague advice that will waste the programmer's time, and on the other hand you don't want to spoon-feed them every single

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Week 11

Monica Dinculescu

It's been another week of people being shit, innit? I know I should be

used to this by now, but it never ceases to amaze me the capability people have for violence

and evil, and the capability the patriarchy has to excuse it. "Dude had a bad day"? For real real?

If that excuse has legs then you'd expect every uterus-holding person to do at LEAST a murder

per year. And yet it's always white dudes who haven't ever thrown up from cramps or had to deal with birth control hormones that feel they're

entitled to attention or sex or whatever and go on these racist murder sprees. And they're

going to continue to as long as the media excuses them as "bad days", or gives them a platform by

encouraging racism and xenophobia. If you're a Fox news anchor who's ever said the

words "china virus" in the last year...these murders are on you pal. I see you, and I deeply hope you don't have a good night of sleep for the rest of your life.

I took all this anger and donated to Red Canary Song and also pointlessly stabbed linoleum with it. The goal is to carve things a computer generates, but at the moment that's a bit too hard for me (I make the computer generate a lot of lines and uhhh that's a lot of cutting), so I've been carving some non-generative stuff. They're all a bit surrealist, because that is extremely my shit, but I don't really know what to do with them.

I also don't understand my audience. For example, I made this computer plant. I thought it was meh, so I wasn't gonna put it up in the store, but after tweeting about it, it ended up selling out in a nanosecond. I made this March Madness basketball beet that's like...actually well carved and in two colours and cute and nobody cared. The same happened with the generative stuff – the chaos attractors I thought were deeply boring sold out instantly, the wings in flight that I absolutely adore are a no-op. I don't get it, y'all.

I am back on my procedural bullshit. Adam has been telling me about the Mentalist for months now, and holy shit: he was right. A++ procedural murders, be back in 7 seasons.

Every year I say I'm going to do a March Madness bracket where I ignore the chalk and just pick it on names and mascots, and every year I think the chalk knows best so I don't and get busted on like day 1. I see you Oral Roberts toothpastes and Loyola wolf boys, I should've kept you going to the final 4 like I wanted to, eh?

Related: defensive fouls in basketball are absolutely bull-shit.

Zach and I are rewatching all of the Marvel movies in story chronological order. This means that I went on several really productive 4 am YouTube holes of watching Avengers bloopers, every recorded interview with Robert Downey Jr, and that whole Ally McBeal season he was on. Imagine if I could put that kind of energy in watching videos about quantum mechanics or something. Instead, I can tell you that he met his wife when filming Gothika, that Tom Hiddleston couldn't stop laughing during that one scene in Avengers 1 when Hulk smashes him into the Stark tower apartment floor, and that Mark Ruffalo is scared of needles so he doesn't have the Avengers tattoo everyone else got. WHY AM I THIS WAY.

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A Brief History of the Polish Winged Hussars

George Levrier-Jones · History in 28 Minutes

The Winged Hussars of the Polish Lithuanian Commonwealth dominated the battlefields of Eastern Europe from the 16th to the 18th centuries. Sporting massive artificial wings on their armor, these aristocratic warriors were among the most feared and cohesive cavalry forces in European history.

Eastern Europe has a deep equestrian and cavalry tradition, due in part to its topography. Active from the 16th to the 18th century, the Winged Hussars showed a consistently remarkable ability to seemingly defy the odds snatching victory from otherwise imminent defeat.

Eastern Europe has a rich history and tradition of elite horsemanship. Much of the region is comprised of plains and flat grassland advantageous to horse rearing. Successive waves of elite horse warriors such as the Huns, Magyars, and Tartars had raided and settled the area at various times. Likewise, much of the warfare that defined this region throughout the Middle Ages and early modern era involved larger cavalry forces than the more heavily populated and more urbanized parts of Western Europe.

Hussars derive their name from the Hungarian “Huserones” and originated in the Kingdom of Hungary sometime in the late 15th century as a light cavalry force. The name later became incorporated by a whole host of Central and Eastern European countries such as Serbia, The Holy Roman Empire and Poland becoming renowned as some of the best light horse units in all of Europe.

The Polish Lithuanian Commonwealth was one of the premier states in Early Modern Europe. Engaging with a host of enemies mainly the Ottoman Turks, Swedes and Russians, a series of drastic military reforms were undergone to bolster and safeguard its territory and people as the state grew and became more vulnerable. These included but were not limited to modernizing land and naval forces and modifying its already prolific cavalry.

Drawing on the litany of pre-established equine traditions, the Winged Hussars as they became known drew their ranks from the upper classes and nobility and adapting their role into a heavy shock cavalry instead of the more traditional nimble light horse tactics of their predecessors.

Riding in tight formation these heavily armored horsemen could smash into enemy formations with lethal efficiency thus enabling infantry and lighter cavalry units to exploit the gaps and gain tactical advantages ensuring victory. The Winged Hussars displayed impressive capability in the diverse geographical and climatic conditions from which they operated.

The Winged Hussars were well armored, well equipped and rode expertly bred war horses. One reason in which their ranks were filled almost exclusively by nobles was the sheer cost of such expenditures.

Wings: The signature and iconic wings worn by the Hussars were fashioned out of feathers from a variety of raptor birds, mainly Eagle and Falcon. Originally the wings were placed on their saddles before being fastened onto the backs of their armor. The purpose was twofold. In an era of extravagant military arms and uniforms the Wings were visually impressive on parade grounds and intimidating foes. The Wings also may have produced an odd and terrifying din when the Hussars charged, frightening enemy soldiers and horses in the process.

Armor: Hussars were clad in heavy steel armor typical of the era and often sported exotic animal furs such as leopards and saddles decked out with fine silk and lace. Beneath their armor and hides Hussars typically donned red or crimson short coats called zupans with helmets encrusted with gems or plumed with exotic feathers. The Winged Hussars took great pride in their appearance on and off the battlefield.

Horses: Expensive warhorses typically Polish-Arabian breeds were the predominant mount used by the Winged Hussars. These horses were both incredibly strong as they could carry a heavily armored rider in addition to their great endurance with long marches ranging across a variety of terrain in and around Central and Eastern Europe.

Weapons: They were well trained, well-armed and capable of wielding a lance, (kopia) saber, (szabla) and various firearms with expert lethality. The Hussars could combat a variety of foes. From conventual infantry and cavalry units which defined 16th and 17th century European armies to the expert light cavalry of the Tatars.

Khlushino: One of the finest victories performed by the Winged Hussars was the Battle of Klushino fought on July 4th, 1610. The heavily outnumbered Polish-Lithuanian force of about 6,000 with the bulk of the army of 5,000 Winged Hussars crushed a Muscovite army of over 30,000. Over the course of a five-hour battle the Winged Hussars superior training, tactics, and troop cohesion made it possible to smash their adversaries’ lines, disintegrating their forces and routing them in the process.

Vienna: The Winged Hussars most renowned moment came in September 1683 at the Siege of Vienna. For weeks the Ottomans under Mustapha Pasha had been battering the “Golden Apple of Europe” coming closer to victory with each passing day. Finally, a coalition of predominantly Catholic Holy League Forces arrived to lift the Siege. John Sobieski King of Poland an experienced soldier led a contingent of roughly 4,000 Winged Hussars. Devising a plan Sobieski outflanked the Ottoman Camp and along with his Winged Hussars led what was possibly the largest cavalry charge in the history of Europe at the head of some twenty thousand horsemen devastating the Ottoman expeditionary force and relieving the city in the process.

The site has been offering a wide variety of high-quality, free history content since 2012. If you'd like to say

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A U.S. Civil War Spy in Europe: Henry Shelton Sanford

George Levrier-Jones · History in 28 Minutes

Who does not enjoy a good spy story? The Civil War, though fought on American soil, was also waged in drawing rooms, chancelleries, and counting houses across Europe. In that shadow war, few figures were more important—or more obscure—than Henry Shelton Sanford.

Sanford was not the kind of man one would cast as a master spy. He did not resemble the polished, worldly intelligence officer of fiction. That was precisely why he was effective. Born in Connecticut to a prosperous family whose wealth came from manufacturing brass tacks, Sanford grew up comfortably connected. One of his ancestors had served as governor of the state. He attended Trinity College and studied in Germany, though he never graduated from either. What he lacked in formal credentials he made up for in money, mobility, and social access.

At just twenty-four, Sanford entered diplomacy, appointed secretary to the American legation in St. Petersburg in 1847. A year later he moved to Frankfurt, and in 1849 to Paris, where he remained for five years, eventually rising to *chargé d'affaires*. In 1861 Abraham Lincoln named him minister to Belgium. His official portfolio included trade agreements, naturalization treaties, and consular arrangements such as the Scheldt Treaties of 1863, which governed customs duties and navigation rights on one of Europe's most important commercial waterways. But Sanford's formal responsibilities were the least important part of his job. His real assignment was counterespionage.

What made Sanford valuable was not diplomacy but deniability. He was wealthy enough not to require a salary, socially connected enough to travel freely without raising suspicion, and unburdened by the technical minutiae that tied other diplomats to their desks. Like many ministers of the era, he was assumed to be a gentleman abroad—sightseeing, attending receptions, and occasionally reporting home. That assumption was his camouflage.

In reality, Sanford was one of the principal architects of the Union's covert war in Europe. Secretary of State William H. Seward entrusted him with authority far exceeding his nominal rank. Sanford was permitted to travel freely across the Continent and into Britain. He was given access to a secret fund of roughly one million dollars—a staggering sum at the time—to finance intelligence gathering, influence, and interference. His mission was straightforward to describe and extraordinarily difficult to execute: prevent the Confederacy from acquiring ships, weapons, credit, and diplomatic recognition.

Jefferson Davis and his government understood that they could not prevail in a prolonged war without foreign assistance. The American Revolution provided the model: French intervention had transformed rebellion into victory. Confederate leaders hoped Britain or France might play a similar role in 1862. Short of recognition, they needed

rifles, cannon, powder, ships, and financing—resources Europe could supply in abundance if the Union blockade could be breached.

Seward and Charles Francis Adams, the American minister in London, formed the official diplomatic front. Sanford was tasked with the unacknowledged work behind it. From Brussels, Paris, and London, he assembled a private intelligence service. In Britain, he employed a police detective who ran operatives in major ports and industrial centers. Shipyards, foundries, arms manufacturers, insurers, and brokers were watched closely for signs of Confederate activity.

Identifying Confederate agents was rarely difficult. They were Americans from the seceded states, often with unmistakable accents and known loyalties. Some were serving Confederate officers; others were businessmen acting as intermediaries. The challenge was not knowing who they were, but discovering what they were doing.

In a world without telephones or secure communications, conspiracies traveled on paper and wire. Letters moved through the post. Contracts were telegraphed. Shipping instructions passed between offices and ports. Sanford targeted all of it.

Sanford's agents bribed postal workers to copy or intercept Confederate correspondence. Telegraph clerks were paid to divert or decode messages. Clerks inside factories and shipyards were induced to hand over specifications, contracts, and delivery schedules. Couriers carried intelligence between Belgium, France, and Britain. At times, Sanford simply "borrowed" Confederate letters long enough to read them before returning them to circulation.

Through business contacts, he tracked cotton shipments, arms purchases, and financial transactions. When necessary, he quietly pressured European firms not to deal with the South. The aim was not dramatic disruption but steady suffocation.

Two Confederate operatives were of particular importance: Caleb Huse and James D. Bulloch. Huse, a West Point-trained officer and former chemistry instructor, served as the Confederacy's principal arms buyer. Operating across Britain, Austria, Prussia, and beyond, he negotiated most of the weapons contracts that eventually supplied Southern armies. Bulloch oversaw naval procurement, including the construction of commerce raiders in British shipyards. One of them—the Alabama—would devastate U. S. merchant shipping before being sunk off Cherbourg.

Sanford tracked both men closely. He fed intelligence to Adams in London, worked to delay or derail their transactions, and ensured that Washington knew when ships were likely to sail. When formal channels failed, less formal methods were sometimes employed. As Sanford joked to Seward in one letter, "accidents are numerous in the [English] Channel, you know."

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David Livingstone - An Inspiring Victorian

George Levrier-Jones · History in 28 Minutes

David Livingstone stands as one of the most celebrated figures of the Victorian age, a missionary, explorer, and abolitionist whose name became synonymous with Africa's vast, unmapped interior. Born on the 19th of March, 1813, in Blantyre, Scotland, Livingstone's early life was one of humble beginnings. The second of seven children, he grew up in a small tenement room above a cotton mill where his father worked as a tea salesman and Sunday school teacher. From the age of ten, Livingstone himself worked twelve-hour shifts at the mill, his small wages helping to support the family. Yet even amid such hardship, he displayed an unrelenting thirst for learning, studying Latin and theology late into the night with the aid of a single flickering candle. His self-discipline and curiosity earned him a place at Anderson's University in Glasgow, where he trained in both medicine and theology. It was during this time that he became inspired by the writings and appeals of the London Missionary Society (LMS), whose vision of combining medical work with Christian mission would become the cornerstone of his life's endeavor.

In 1840, Livingstone was ordained as a missionary doctor under the LMS and sailed for Africa, a continent largely unknown to Europeans beyond the coastal regions. His first posting was in the Bechuana country (modern-day Botswana), where he worked alongside the veteran missionary Robert Moffat. There, Livingstone quickly distinguished himself not only for his medical skills and fluency in local languages but also for his belief in establishing missions far inland, away from European colonial influences. His early travels introduced him to the harsh realities of African geography and the challenges of crossing vast deserts such as the Kalahari. Livingstone's marriage to Moffat's daughter, Mary, in 1845 marked the beginning of a partnership often tested by the dangers of exploration and illness.

Livingstone's first great achievement came in 1849 when he crossed the Kalahari Desert to reach Lake Ngami, a body of water previously unknown to Europeans. His reports of this journey captured the imagination of the British public, eager for tales of adventure and discovery. Determined to find new routes for legitimate trade as an alternative to the brutal slave routes that scarred the continent, Livingstone pushed further north. Between 1851 and 1856, he traversed thousands of miles, becoming the first European to cross the African continent from west to east. His expedition from Luanda on the Atlantic coast to Quelimane on the Indian Ocean was a feat of endurance that won him worldwide fame.

It was during these years that Livingstone made one of his most famous discoveries: the great waterfall on the Zambezi River, which he named Victoria Falls in honor of Queen Victoria. The native name, Mosi-oa-Tunya—"The Smoke That Thunders"—he preserved in his writings, noting its grandeur and spiritual significance to local peoples. His detailed journals and maps from this period

were meticulously kept, later forming the basis for his book *Missionary Travels and Researches in South Africa* (1857), a major publication that enthralled readers and established his reputation as both a scientist and a man of faith. The Royal Geographical Society awarded him its gold medal, and his observations contributed significantly to the European understanding of African geography, geology, and ethnography.

Livingstone's later expeditions, particularly the Zambezi Expedition (1858–1864), were less successful but no less ambitious. Appointed by the British government to explore the navigability of the Zambezi River and its tributaries, he hoped to open up routes for trade and Christian missions that would undermine the slave trade. However, the journey was plagued by disease, logistical failure, and tragedy, including the death of his wife Mary from malaria in 1862. Despite these setbacks, his scientific work remained meticulous. He recorded flora, fauna, and mineral deposits, and his notebooks, many of which survive in archives such as the National Library of Scotland bear witness to a disciplined observer driven by both humanitarian and scientific motives.

In the later years of his life, Livingstone became increasingly preoccupied with finding the source of the Nile, a mystery that had fascinated explorers for centuries. His travels took him deep into Central Africa, where he lost contact with the outside world for several years. Rumors of his death circulated widely in Europe until, in 1871, the Welsh-born American journalist and explorer Henry Morton Stanley was dispatched by the *New York Herald* to find him. Stanley's long and arduous search ended in the town of Ujiji on the shores of Lake Tanganyika, where he greeted the weary, bearded missionary with the now-legendary words, "Dr. Livingstone, I presume?"

The meeting between Livingstone and Stanley became one of the most famous encounters in exploration history. Livingstone, though weakened by illness and years of hardship, was still resolute in his mission. Stanley, impressed by the older man's determination and moral conviction, provided supplies and encouragement. The two men explored parts of Lake Tanganyika together before Stanley returned to the coast with news that Livingstone was alive. Stanley's own life, though often overshadowed by this single encounter, was remarkable. Born John Rowlands in Denbigh, Wales, in 1841, he endured a harsh childhood before emigrating to the United States, where he served as a soldier, sailor, and journalist. His transformation into Henry Morton Stanley came after being adopted by a wealthy merchant of that name. His later explorations, including the charting of the Congo River, would establish him as one of the most controversial and driven explorers of the 19th century.

David Livingstone, however, never returned home. His final years were spent in relentless pursuit of the Nile's source, often under conditions of extreme suffering. His final journals, preserved on fragile paper and sometimes written in hasty jives when ink ran out, reveal

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African American Servicemen in Britain During World War 2

George Levrier-Jones · History in 28 Minutes

Just after 9pm on a cool September evening in 1943, a large group of soldiers calmly walked the mile home to camp where they armed themselves with tommy guns, ammunition and bayonets. Putting themselves into formation, they marched back into town, three-a-breast. The sound of their army-issue boots striking the road for nearly a mile echoed heavily in the pitchy-ink of the blacked-out night-time and is something witnesses remember to this day. It seemed as if a 'whole company' of troops was moving through the night, it was said later.

In England, Major Charity E. Adams, Columbia, South Carolina., and Captain Abbie N. Campbell, Tuskegee Institute, Tuskegee, Alabama, inspect the first members of the African American Women's Army Corps assigned to overseas service.

Undoubtedly the troops who were on the move were ready for the fight of their lives – it just wasn't the official enemy they had in their sights. Because unbelievably this was not happening in mainland Nazi-occupied Europe, but on Britain's homefront – specifically the market town of Launceston in Cornwall. And these were American soldiers. Military police patrolling the town could sense impending danger. 'Everything was so tense that evening that we thought that something might start,' said one. Another added that all evening '...you could feel the tenseness in the air.' Even publicans working in the town's many drinking houses felt this was the calm before the storm. One shut early that evening saying how he just sensed '...something brewing.'

Suddenly the marching troops appeared 'in a body' from out of the darkness to encircle a group of military policemen, fellow Americans, who were standing chatting next to a jeep parked near the town's war memorial. 'We saw forty to fifty soldiers coming up the street. They had overcoats on. They walked up almost in formation, and straight toward us... and [we] thought trouble was about to begin,' said one of the surrounded. A man, who seemed to be spokesman for the group, said very quietly: 'Why don't you let us come into town, come into the pubs?'. Flashlights snapped on. 'Hands up!' was shouted. The military police raised their arms and backed up. As they did, 'I heard bolts open on rifles,' said the jeep's driver. There was just time for the terrifying realisation to sink in that their compatriots were not only armed but already taking aim when: 'I heard a bolt crack and a shot landed at our feet. Someone hollered 'DUCK'. I jumped in behind the wheel of a jeep.' Next, a volley of fire. 'I felt a bullet whizz past me.' A flashlight revealed a soldier 'with a denim hat and overcoat firing a rifle from the hip and he was really pumping them out.' A pause. Then chaos as British soldiers, civilians, WAAFs and Land Army girls, as well as the Americans under fire, scrambled for cover amid ricocheting bullets. One old man told the Daily Mirror the next day: 'There hasn't been anything like this since the days of the smugglers.'

No-one knows for sure exactly how many soldiers were armed and fighting that night. What is universally acknowledged is a large number was involved – from the 581st Ordnance Ammunition Company who were firing at soldiers from the 115th Infantry's Second Battalion. It was all over in five minutes before the shooters melted away into the night. What they left behind was a shot up town centre, soldiers and citizens shaken, store windows in shatters, two hole-ridden US army jeeps (it subsequently took 20 soldiers to lift them bodily away), two sergeants with mashed-up legs, the visiting US army with its reputation hugely-dented and bullet holes in Cornish bricks and mortar which for more than seventy years were the sole reminder of an all-American gunfight army authorities wanted forgotten and tried their best to obscure. Because the inconvenient truth here was that these were members of an African American ordnance company who were taking on the white soldiers who policed them. The level of injuries given the firepower on hand that night shows precisely that wholesale slaughter was certainly not the intent, although military prosecutors defied their own investigators recommendations and insisted on bringing attempted murder charges alongside mutiny et al. The 'mutineers' were making a point and it was one that was needed to be made.

There were around 130,000 African Americans among the 1.5 million US servicemen who were in the United Kingdom at any one time in World War Two – altogether 4 million Americans would come to Britain. But this segregated army had an inherent racial friction which began to spill over into violence with increasing frequency whenever the two races met in Britain's 'green and pleasant' land. Riding on the tide of simmering racial tension in US training camps and explosive riots in five American cities during the long hot 'bloody' summer of 1943, this enmity inevitably floated across the Atlantic with each wave of arriving servicemen.

At first it baffled the British. Despite ruling an empire upon which 'the sun never set' there were surprisingly few people of colour, roughly 15,000, in Britain during World War 2. Undoubtedly, in such a mono-cultural society, racism was bound to thrive as proved by race riots in 1919 and exemplified by the experiences of Learie Constantine, the West Indian cricketer, who came to live in Britain in the 1920s and described how 'personal slights' were 'an unpleasant part of life in Britain for anyone of my colour.'

But evidence from thousands of censored letters, secret reports from the Ministry of Information's Home Intelligence division, surveys for Mass Observation (the nascent polling organisation) as well as editorials and letters to newspapers and government departments shows this confusion amongst ordinary Brits soon morphed into outright rejection of the 'colour bar' – and decided support for African American troops. From George Orwell who kicked off his first article for Tribune with "The general consensus of opinion seems to be that the only American soldiers with decent manners are the Negroes" to a Blackpool factory worker raging against how "...the American troops literally kick, and I mean kick, the coloured soldiers off the pave-

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Mock The Location Href Property

Gleb Bahmutov

Let's say the application changes the URL using the assignment:

```
1 location.href = 'https://acme.com'
```

Can we prevent the URL change? For example, we might want to limit our test to the current origin. Let's do it using the following example page:

```
index.html 1 2 3 4
```

```
Hello World
```

```
app.js 1 2 3 4 setTimeout ( () => { console . log ( 'changing window location to acme.com' ) location . href = 'https://acme.com' }, 1000 )
```

The initial Cypress test does not prevent the navigation to "acme.com"

```
cypress/e2e/spec.cy.js 1 2 3 4 5 6 7 it ( 'sets the location HREF' , () => { cy . visit ( 'index.html' ) cy . contains ( 'h1' , 'Hello World' ) /\ confirm but do not allow the application / \ to navigate away to the new URL /\ Tip: app sets it using "location.href = ..." command }
```

☒ The source code for this blog post is located in the repo bahmutov/with-window .

I have recorded a short video going through this example and my solution

Cannot stub Location object

We need to prevent location.href = ... assignment by making the property read-only or better: using a custom object property definition with a setter stub function. Unfortunately, we cannot overwrite the location.href property:

Ok, maybe we can mock the entire location object? It is a property of the global window object:

Unfortunately, the location property itself cannot be overwritten either.

We need another way. In the blog post Stub The Unstubable I have shown one possible solution that modifies the application's source code to create an intermediate proxy "Location" instance. The E2E test can control that object. It is imperfect solution, as it requires source code modifications, which might be unavailable.

Let's find another way.

JavaScript with keyword

JavaScript has a pretty obscure operator with that you should definitely NOT use in production, but can help us with our testing needs.

```
1 2 3 with (expression) { ... }
```

The above syntax adds the expression result into the variable lookup chain. For example:

```
1 2 3 4 5 6 7 8 9 let a, x, y; const r = 10 ;
```

```
with ( Math ) { /\ where is PI, cos, and sin defined? a = PI * r * r; x = r * cos ( PI ); y = r * sin ( PI / 2 ); }
```

In the example above, the PI , cos , and sin are properties of the Math object. By using with (Math) we are forcing the browser to look up these identifiers in the Math object (before going up to the window object).

Tip: the with (expression) syntax is hard to read and understand. A simple spread operator would be much more preferable way of coding the above example:

```
1 2 3 4 5 6 let a, x, y; const r = 10 ; const { PI , cos, sin } = Math a = PI * r * r; x = r * cos ( PI ); y = r * sin ( PI / 2 );
```

The E2E test

The application's code accessing the location object is loaded by the resource. Let's wrap this code using a fake window object just so we can "sneak" in a fake "location" object of our creation.

```
cypress/e2e/solution.cy.js 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 it ( 'sets the location HREF' , () => { cy . intercept ( 'GET' , 'app.js' , ( req ) => { req . continue ( ( res ) => { /\ wrap app's code with a fake window object /\ that has overwritten location object res . body = ` const fakeWindowObject = { location: { href: " , }, } with (fakeWindowObject) { ${res.body} ` } } ) . as ( 'appJs' ) cy . visit ( 'index.html' ) cy . wait ( '@appJs' ) cy . contains ( 'h1' , 'Hello World' ) }
```

When the test runs, the application loads app.js and receives the following (modified) script

```
app.js 1 2 3 4 5 6 7 8 9 10 11 const fakeWindowObject = { location : { href : " , }, } with (fakeWindowObject) { setTimeout ( () => { console . log ( 'changing window location to acme.com' ) location . href = 'https://acme.com' }, 1000 ) }
```

The test stays on the same page, but we do see the application printing "changing window location to acme.com".

We need to verify the location.href property really changes to acme.com string. We can put the fake window object on the real window object. Then we can get its value using the cy.window command.

One last tip: the browser caches the app.js resource, thus we need to remove the caching headers in order to receive the full JavaScript source code:

```
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 cy . intercept ( 'GET' , 'app.js' , ( req ) => { /\ delete common cache headers /\ so the browser gets the real app.js source code delete req . headers [ 'if-none-match' ] delete req . headers
```

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2017: another year in review

Monica Dinculescu

You can tell I hate writing year in reviews because this one is really, really late. I tend to hate bragging, and I definitely hate introspective and, in particular, I always think I am underperforming (and that's fine). However, that's usually not true, and writing a year in review forces me to see the awesome things I did, so even if I did end up underperforming, at least I can learn from that. That's the whole point of post-mortems, right?

As usual, here's life as GitHub saw it. Red text is projects I've shipped, black text is conferences I've spoken at. Technically I didn't speak at Blinkon, but I spoke to people at it so hey, counting it.

The reason why this post mortem is important to me is that before writing it, I literally thought this year was bad and I just "did fewer things". But that's not actually true!

I wrote less code than in 2016 (2713 vs 3153 contributions), but that's totally irrelevant? GitHub contributions are a fake idea, and I'm the first one to tell you this, so I don't know why I get worked up about them every year. I contributed to weird branches (which don't get counted), and did a lot of weird explorations that obviously never got merged. I planned things. I wrote design documents. I reviewed design documents. I formed strong opinions. I learnt Redux. Eat it, contributions graph.

I gave fewer talks than in 2016 (7 vs 8), but I enjoyed conferences more. I went to a conference where I only hung out with badass women. I went to Railscamp, where I had no wifi and canoed and wore a headlamp like a giant dork. I MC-ed Chrome Dev Summit with Mariko, which was scary, and intimidating and incredibly fun.

I built more side projects (10 vs 9). One of them got featured in The Verge !

I worked fewer weekends (16 vs 28 days. Goal: 0). And I don't mean on work-work , I mean, at all . On the weekends I play Stardew Valley, or make pottery, or knit, or watch an entire season of Riverdale because I can . Maybe this means I'll work on fewer side-things, and maybe this will hurt my career, but it will keep me happier, not burnt out, and less likely to murder my partner. And that's p important.

I joined the Unicode Emoji committee. YES. Really. It took me a year of emails, and even now I'm pretty sure I'm the least useful member out of the whole bunch, but that's ok, because I can get better!

I became broadly aware of a lot of technical things, but not necessarily deeply aware of them. I am really confident I understand Web Components, emoji and web fonts really well , but I still don't really know how any of our polyfills work, or how to fix a sizeable bug in Polymer, or what to do about async/await and stuff. In the last couple of months of the year I started learning more about http/2 push and link

rel=preload , but I feel it's a broad sort of understanding. I don't know yet what kind of person I want to be: "I understand something really well in all its intimate aspects which makes me an expert", or "I understand many things well enough to have opinions and advice, but not well enough to be an expert in any of them". This is the thing I want to figure out this year.

I learnt how to skateboard. I saw one of my favourite bands in concert. I ordered a coffee entirely in Japanese. I learnt how to make mugs and bowls and bottles, with my hands. I made new friends, and I didn't piss off any of my current ones (that I know of). I nuzzled dogs, and cats, and an otter. I still didn't spoil Star Wars. I turned 32.

I don't do resolutions because they don't really work for me, but I heard a good one from a coworker: do 12 new things next year.

So, I will.

id="">♥

/\ also available: 2016 and 2015 years in review, that were actually on time.

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Cypress Dependencies Through A Docker Image

Gleb Bahmutov

If you are testing a website, the DEV dependencies do not change very often. You might bump Cypress version once in a while, add or upgrade a Cypress plugin , but in general the Node dependencies are changed less frequently than the spec files. Thus running npm ci on every test job is bound to be repetitive work that slows down your testing pipelines.

Of course, each CI allows you to cache dependencies between the jobs - but why do you need to even think about it? You want to cache all dependencies the very first time npm ci runs for the given package.json or package-lock.json file, not the first time each the workflow runs! Most CIs let you control the container image used for running tests; I assume you use one of Cypress Docker images or one of Cypress Docker images with browsers installed . So what if we could create our own Docker image based on Cypress (or any appropriate Docker base image you might want) and run npm ci in that image, and then use that Docker image to simply run our current tests?

I had the idea of caching separately PROD and DEV images inside Docker containers a looooong time ago, see the bahmutov/double-docker repo. In this blog post, I will show a simple approach that works for repositories with tests only, the situation described in the blog post Separate Application And Tests Repos GitHub Actions Setup . We will use GitHub Actions and the public Docker registry.

In the nutshell:

whenever the user changes package.json or Dockerfile , we build a new Docker image with the node_modules and Cypress binary folders (but no other source code)

we push the built Docker image to the registry

every CI run simply pulls that Docker image and checks out the tests into the container

the tests run immediately after the checkout step, since there is nothing to install

Let's see the code

Build and push

Run on GitHub Actions

GitHub Container Registry

☑ I have the working example in the public repo bahmutov/cypress-tests-image .

First, let's look a the Dockerfile

Dockerfile 1

2

pick the image to build from

either the base or the browsers image

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Roman Trade with Modern-Day Sri Lanka

George Levrier-Jones · History in 28 Minutes

When Roman sailors and merchants in the first centuries of the Common Era looked southward from the mouth of the Red Sea, the island they called Taprobane, now almost universally identified with Sri Lanka, appeared in their geographies as a rich and mysterious partner in an expanding Indian Ocean trade network. Classical geographers and travel writers treated Taprobane both as a place of fabulous commodities and as a real staple of long-distance exchange: it appears in the *Periplus of the Erythraean Sea*, in Pliny the Elder's *Natural History*, and on Ptolemy's maps, and these accounts, read together with archaeological finds around Sri Lanka's ancient ports, make a persuasive case that Greco-Roman traders reached the island's shores, if sometimes indirectly, from the mid-first century CE onward.

The short but vivid *Periplus of the Erythraean Sea*, a Greek seaman's handbook usually dated to the mid-first century CE, is the most direct contemporary Roman-period testimony to long-distance navigation in the Indian Ocean. Its anonymous author describes routes, seasons and goods, and although Taprobane receives briefer treatment than the western Indian ports, the *Periplus* places the island within the author's pragmatic commercial map: it is a known source of gems, ivory and other sought-after products and a waypoint for ships that ranged along the eastern Indian seaboard and beyond. The *Periplus* also reflects the adoption of the monsoon crossing, knowledge of seasonal wind patterns credited to figures such as Hippalus, which made regular oceanic linkages between the Red Sea and South Asia feasible and economically attractive for Roman subjects based in Egypt.

Pliny the Elder and Ptolemy amplify the picture of Taprobane as an object of classical curiosity and commerce. Pliny compiled earlier reports and travelers' tales into his encyclopedic *Natural History* and explicitly names Taprobane among the distant islands whose products reached Mediterranean markets; Ptolemy's *Geographia*, meanwhile, institutionalized Taprobane on maps, even if the island's scale and position were distorted in late antique cartography. These literary testimonies outline what the Roman-world readers believed they were buying from the Indian Ocean: not only luxury items such as pearls, gems, spices and exotic woods, but also useful commodities—elephants for war and ivory for luxury works—that made a voyage to Taprobane, or through agents who dealt with it, commercially worthwhile.

Textual notice must be matched to material traces, and here archaeology supplies concrete, if sometimes contested, evidence. Excavations and survey work at Sri Lanka's ancient trading sites—most notably Mantai (the harbor complex at Mannar), Godavaya and certain finds around the southern ports and the Ruhuna region—have produced imported ceramics, Indo-Roman wares, and metal finds datable to the first few centuries CE. These finds indicate

that goods and perhaps persons moved between the island and the wider Indian Ocean economy at the time classical authors wrote. Archaeologists have recovered Roman or Roman-style amphora sherds and Mediterranean imports mixed with local and South Indian ceramics in contexts that often line up with the mid-first through third centuries CE, supporting the textual evidence that Taprobane was part of long-distance trade networks rather than an isolated curiosity.

Numismatics has contributed one of the most headline-grabbing types of evidence: Roman and Indo-Roman coins found on the island and in neighboring Indian ports. Scholars have catalogued a variety of coins—official Roman denominations, later Byzantine and Sasanian pieces, and locally struck imitations, that turn up in hoards and stray finds across Sri Lanka's archaeological record. The presence of Roman gold, silver and bronze specimens, and of imitations that adapt Roman portraiture and types, demonstrates multiple channels of contact: direct importation, trade mediated by South Indian merchants, and the island's participation in a wider monetary ecology that incorporated foreign coinage as bullion or prestige money. At the same time, numismatists caution that coin deposits are slippery evidence for regular commercial routes, coins travel, are hoarded, reused and sometimes arrive centuries after they were minted, so they illuminate connectivity without always proving direct, continuous Roman state involvement.

Material culture beyond coins, such as fragments of Mediterranean amphorae, glassware and certain classically styled objects helps round out the picture. Amphora sherds linked by form and fabric to Mediterranean production argue that Mediterranean foodstuffs (wine, preserved fish or oil) or at least their containers were carried into the Indian Ocean system. Yet many of the "Roman" finds in South Asia were funneled through intermediary entrepôts on the Malabar coast and the Arabian littoral; the principal agents of exchange in this era were often Indian, Arabian and later Southeast Asian merchants who operated the coastal networks and transshipped cargoes to and from Sri Lankan anchors. Consequently, although Roman subjects and merchants surely appear among the actors of this commerce, especially in Egyptian port cities like Berenice and Myos Hormos—the day-to-day movement of goods to Taprobane seems to have been largely run by regional middlemen who connected Mediterranean markets to South Asian and island producers.

Certain Sri Lankan exports made the island especially prized by Mediterranean consumers. Classical authors and later commentators repeatedly emphasize cinnamon (whose precise identification in ancient texts is debated), pearls from the island's offshore banks, high-quality gemstones and ivory. Archaeology confirms the island's role in pearl fishing and its access to precious stones and fine timber, while epigraphic and local traditions record long-standing maritime commerce. The intersection of demand in the Roman world for luxury consumables and Sri Lanka's

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For Southern Antagonists in the Civil War, a Kindred Desire for Peace Goes Awry

Jesse George-Nichol · HistoryNet

On December 18, 1860, John J. Crittenden of Kentucky introduced a compromise plan to the U. S. Senate. Just two days later, South Carolina would become the first state to secede from the Union, and within six weeks, six more Southern states would follow suit. But while Dixie fire-eaters were driving their states pell-mell toward disunion, Senator Crittenden and other moderates were working to broker a sectional adjustment — one that could, they hoped, soothe Southern fears about Abraham Lincoln's election and stay the secession tide in the South.

The Crittenden Compromise would be central to these efforts during the winter and spring of 1860-1861. It represented an attempt to settle the slavery question once and for all, drawing on the tradition of grand settlements like the Missouri Compromise and the Compromise of 1850. Indeed, the cornerstone of Crittenden's plan was a constitutional amendment that would divide the remaining Western territories along the old Missouri Compromise line, barring slavery above and protecting slavery below the 36° 30' parallel.

Moderates like Crittenden hoped that this might be enough to secure the loyalty of the remaining Southern states to the Union. This might, in turn, make Republicans more willing to let the secession crisis play out, and it might eventually make the seceded states more willing to return to the Union. Yet most Republicans, including Lincoln, refused to countenance any further extension of slavery into the territories. Attempts by moderates to push through the Crittenden Compromise repeatedly foundered against this opposition.

Compromisers struggled, too, against the opposition of Southern secessionists, who argued that it did not do enough to protect slavery from the threat of an empowered Republican Party. Over the course of the secession crisis, it became clear that the leaders of the seceded states had no interest in negotiation or returning to the Union. Southern rights advocates in the states that had not seceded also complicated the project of compromise; their demands for more concessions meant there was no consensus around Crittenden's or any other compromise measure even in those states.

Moderates attempted to push through the Crittenden Compromise, but met opposition from both sides.

One such antagonist was Virginia's John Robertson, a prominent Democrat and judge from Richmond. The state legislature sent him as a commissioner to the seceded states in early 1861, and he returned with assurances of the new Confederate States' sympathies with Virginia. They are "bone of her bone, and flesh of her flesh," he reported.

The outbreak of fighting at Fort Sumter on April 12 provided the push for which many Southern hardliners had been

hoping. Abraham Lincoln responded by issuing a call for 75,000 troops to put down the rebellion in the South, and in short order Virginia, Arkansas, North Carolina, and Tennessee seceded and joined the new Confederacy. The start of the war seemed to signal triumph for militants like Robertson and disaster for moderates like Crittenden. Yet neither man would accept this as the outcome of his labors.

Robertson wrote a letter to Crittenden near the end of April that highlighted just how uncertain the future appeared in that moment. Robertson refused to believe that the collision at Fort Sumter necessarily meant war —and rejected, moreover, the idea that war would accomplish the ends of either side in the conflict. He thus suddenly and unexpectedly found his own goals aligned with Crittenden's, and Robertson begged the old Kentuckian to renew his efforts at conciliation.

From Robertson's point of view, civil war did not seem inevitable, even when armies were massing on both sides of the Mason-Dixon Line. The situation represented a dramatic escalation, to be sure, but in the context of the decades-long sectional crisis over slavery (one that had at other points erupted in violence), observers like Robertson could imagine outcomes other than intestine war.

John Robertson to John J. Crittenden, April 28, 1861

Dear Sir,

No man could have more earnestly striven than yourself to [resolve] the feuds, whose increasing fury, already advanced to the stage of murderous conflict, threatens to involve thirty millions of men in the horrors of civil war. However I may have differed with you, looking from a Southern view, as to the acceptability of the terms of adjustment you proposed, I never doubted that you regarded them as just, or, at least, as preferable to the evils otherwise to ensue, and as the best which could possibly be obtained. The event has proved that, moderate as they were, the ruling faction [the Republican Party] would be content with none but such as would degrade the South. Wellnigh desperate is the condition to which that faction has reduced this country. The fact now stares them in the face that the Union is dissolved beyond all hope of restoration, at least, in our day. Yet they are threatening to preserve the Union by force. They read the riot act to millions of men, nay, to sovereign States, who are to be coerced into friendship by their foes at the point of a bayonet. But, waving all recrimination, not insisting on the absurdity of the idea, or the impossibility of reducing the South to an ignominious submission, or the certainty that their subjugation, if possible, would defeat the very object their enemies profess to desire (namely, the preservation or restoration of the Union), by converting States into vassal provinces (in that character alone can they remain or enter into it), let us inquire if there are no means by which the anticipated consequence of our family jars (now an accomplished fact), the separation of the States, may be recognized by the ruling faction at Washington, without deliberately repeating the most atrocious crime, and steeping their hands still deeper in the blood of

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Turning a childhood interest into a career path with MIT Open Learning resources

Lauren Rebecca Thacker | MIT Open Learning · MIT News Education

Hinata Yamahara was 10 years old, on a family trip to Osaka, Japan, when he started riding the train alone. Those train rides sparked an interest in something he didn't have words for at the time — how cities are built, and how people navigate them.

Now at 17 and applying to colleges, Yamahara has made the connection that investigating cities and how they work is a foundation for urban planning. MIT OpenCourseWare, part of MIT Open Learning, helped him develop knowledge and deepen his passion for walkable, community-oriented urban spaces.

At the time of his first, fateful Osaka train ride, Yamahara lived in Atlanta, after having spent most of his elementary school years in rural Tennessee. Before that, he and his family had lived in Los Angeles. In all those places, he says, it was difficult for kids to get around. In fact, getting around without a car could be a challenge for anyone.

Whenever he visited family in Japan, he was in awe of the efficiency of the train system and the ease of learning how to navigate it, in addition to the mixed-use buildings that combined commercial and residential spaces. As a 10-year-old, he simply thought they “were really cool” and quite different from what he saw at home. Now, he reflects on the efficient use of space in a country that is much smaller than the expansive United States.

Back at home, Yamahara continued to reimagine the spaces around him, often printing out satellite imagery and drawing new possibilities over what existed.

“Maybe instead of a giant parking lot, I’d draw a subway station and connect it to another part of the map,” he says. “But it was all imaginary. I didn’t know what was possible in terms of regulations or looking at past examples. So, I started to do research and kept coming across OpenCourseWare.”

He remembers a moment of confusion, followed by one of excitement: “What is OpenCourseWare?” and, “Wait, everything is free!?”

OpenCourseWare, an open publication of course materials from across the MIT curriculum, allows users to browse content at their own pace. Learners can watch video lectures, read course notes, and hear from faculty experts, with no enrollment fees or start dates.

Yamahara dove into the content, starting with 11.001J (Introduction to Urban Design and Development). He has since accessed a variety of courses and counts 11.948 (Power of Place: Media Technology, Youth, and City Design and Development) and 11.304J (Site and Infrastructure Systems Planning) among the most memorable.

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Exploring OpenCourseWare resources gave him a strong foundation to take his interest in urban planning and redesign into the real world. In summer 2025, Yamahara completed an internship with an Atlanta-area real estate firm working on a redesign project with a city agency.

“Right now, this city has a downtown area with a lot of parking lots and empty retail spaces. I’m getting to learn firsthand how to develop it into something else,” he explains. “I’m part of meetings where we brainstorm how to create a community feeling, how to make a space creative and walkable, and how to make it somewhere people really want to be.”

According to Yamahara, the experience draws on his appreciation for Japanese efficiency, as well as American inclusivity and his deep knowledge of zoning regulations, community-centered design, and transit equity — all gained through OpenCourseWare materials and resources.

As Yamahara looks toward college and his future, he says he sees two paths influenced by MIT’s free educational resources. On one path, he studies urban planning, as OpenCourseWare materials have only reaffirmed his passion for the field and given him knowledge and confidence. On another path, he imagines using his time as an undergraduate to explore other interests, including aviation and real estate. On that path, OpenCourseWare would allow him to continue his urban planning education independently.

He’s grateful that he could take advantage of OpenCourseWare resources without submitting an application and considering tuition and financial aid. All he needed was an interest and a determination to learn, and he had both.

“OpenCourseWare has helped me grow from a kid with questions into a student designing solutions,” Yamahara says. “I still ride the train in the U. S. and Japan, but now, I bring a vision with me, too.”

To explore additional lifelong learning offerings from MIT, visit MIT Learn .

standard-article(title: [Episode 115 (A Brief Eulogy for a Commercial Radio Station)], author: [The Memory Palace], source-name: [The Memory Palace], images: (), paragraphs: ([The Memory Palace is a proud member of Radiotopia , from PRX, a curated network of extraordinary, story-driven shows.], [Keep by Nils Frahm], [She cycles through: Where or When by Hal Kemp & His Orchestra, Smarty (You Know it All) by Fats Waller, Dear Mr. Gable: You Made Me Love You by Judy Garland, Hellhounds on My Trail by Robert Johnson, and The Big Apple by Tommy Dorsey and his Clambake Seven.], [Learn about your ad choices: dovetail.prx.org/ad-choices],), insert-map: (:), word-count: 103, edited-for-length: false, debug-mode: false,)

standard-article(title: [Tackling the Awkward Squad for Reactive Programming], author: [Lambda the Ultimate], source-name: [Lambda the Ultimate], images: (), paragraphs: ([https://2020.ecoop.org/details/ecoop-2020-papers/19/Tackling-the-Awkward-Squad-for-Reactive-Programming-The-Actor-Reactor-Model], [Sam Van den Vonder, Thierry Renaux, Bjarno Oeyen, Joeri De Koster, Wolfgang De Meuter], [Reactive programming is a programming paradigm whereby programs are internally represented by a dependency graph, which is used to automatically (re)compute parts of a program whenever its input changes. In practice reactive programming can only be used for some parts of an application: a reactive program is usually embedded in an application that is still written in ordinary imperative languages such as JavaScript or Scala. In this paper we investigate this embedding and we distill “the awkward squad for reactive programming” as 3 concerns that are essential for real-world software development, but that do not fit within reactive programming. They are related to long lasting computations, side-effects, and the coordination between imperative and reactive code. To solve these issues we design a new programming model called the Actor-Reactor Model in which programs are split up in a number of actors and reactors. Actors and reactors enforce a strict separation of imperative and reactive code, and they can be composed via a number of composition operators that make use of data streams. We demonstrate the model via our own implementation in a language called Stella.],), insert-map: (:), word-count: 199, edited-for-length: false, debug-mode: false,)

standard-article(title: [Episode 150: Late One Night], author: [The Memory Palace], source-name: [The Memory Palace], images: (), paragraphs: ([The Memory Palace is a proud member of Radiotopia , a collective of independently owned and operated podcasts.], [A note on shownotes. In a perfect world, you go into each episode of the Memory Palace knowing nothing about what’s coming. It’s pretentious, sure, but that’s the intention. So, if you don’t want any spoilers or anything, you can click play without reading ahead.], [Anyway...], [We start with 666 from the score to The Omen.], [Hear some of Leonard Bernstein’s score to Rebel Without a Cause .], [Hit Denmark by the Portland Cello Project], [And hear some of Musette’s On the Green Tape .], [Bobby’s story is covered in a lot of histories of one hit wonders and novelty tunes. But, if you want to really dig in, you should read, Monster Mash: Half Dead in Hollywood, his autobiography.], [Learn about your ad choices: dovetail.prx.org/ad-choices],), insert-map: (:), word-count: 159, edited-for-length: false, debug-mode: false,)

standard-article(title: [Week 17], author: [Monica Dinulescu], source-name: [Monica Dinulescu], images: (), paragraphs: ([I’ve been skipping weeks because a) literally nothing happens and b) I don’t have a good system to update these notes. They’re a markdown file on a GitHub repo, and I kind of need a computer to edit it, but I also kind of don’t open my computer that much these days? I also keep forgetting which day is Monday. Room for improvement.], [I wrote a blog post about how I generated some tree rings in JavaScript and then carved them as a linocut. It doesn’t actually contain any JavaScript, but it does have a lot of pretty images.], [The last normal thing I did before the panini started was go to Japantown and stock up on apocalypse supplies (snackos, mucho ramen, milk tea powder, korean face things). The first normal thing I did with my 1 vaccine shot was go to Japantown and restock all the things. This wasn’t on purpose, but I am pleased with the serendipity. It is cherry blossom season, so it was very pretty, HOWEVER, some racist asshole vandalized two of the oldest cherry trees there in January. Literally chopped down all of the branches, one at a time. What the actual fuck.], [If you’re against plastic (why aren’t you?) and use deodorants, Dove has started selling refillable ones. I got mine from Target. The refill itself still comes in plastic, but overall it’s far less plastic than the obnoxious amount the normal ones have. I’m excited about this not because this deodorant is particularly amazing, but because Dove is a HUGE brand, and having mainstream brands start looking into more reusable, less-plastic products is a small but exciting progress.], [Speaking of waste, Zach bought us me the most amazing thing: a foodcycler !!! We’ve been composting for years, but I’ve recently started getting lazier about putting the compost back in the freezer when I’m done with it, so our idiot dog has been stealing a lot of compost (which is full of bad things for her like coffee grounds, onion peels, literally half a spaghetti squash rind she ate and threw up for 8 hours). This foodcycler thing takes the compost and dehydrates it and grinds it and in 4 hours is done and gives you back fertilizer at like a tenth of the original volume. We’ve had it for a week and it’s honestly THE MOST innova-

The first normal thing I did with my 1 vaccine shot was go to Japantown and restock all the things.

Monica Dinculescu

standard-article(title: [Episode 184: Betty Robinson], author: [Nate DiMeo], source-name: [The Memory Palace], images: (), paragraphs: ([The Memory Palace is a proud member of Radiotopia from PRX. Radiotopia is a collective of independently owned and operated podcasts that's a part of PRX, a not-for-profit public media company. If you'd like to directly support this show and independent media, you can make a donation at Radiotopia.fm/donate.], [A note on notes: We'd much rather you just went into each episode of The Memory Palace cold. And just let the story take you where it well. So, we don't suggest looking into the show notes first.], [Pollen by H. Takahashi and, later, Photosynthese.], [Wiffle Ball from Joel P. West's score for Short Term 12], [The title theme to Cani Arrabbiati], [Drunken Aviator by the (great, truly) Ida.], [If you are looking to read more about Betty, I'd suggest Rosanne Montillo's terrific book, Fire on the Track.],), insert-map: (:), word-count: 178, edited-for-length: false, debug-mode: false,)

standard-article(title: [Week 18], author: [Monica Dinculescu], source-name: [Monica Dinculescu], images: (), paragraphs: ([Ya look at that? It's May already. My dog's is turning 2 on the 14th, does anyone have any good dog cake recipes or should I just give her an almost-empty jar of peanut butter with the same success rate?], [I got my second fouchie outchie! (#teampfizer). I wouldn't say it bodied me the next day, but I felt uncomfortable enough to sit in bed and read not one but TWO Bridgerton novels. I've now read 1-4 and I think I'm done for a while.], [Speaking of books, I'm finishing up On Earth We're Briefly Gorgeous (Ocean Vuong; reads like poetry but isn't. Review next time), and I'm about to start up The Vanishing Half (Brit Bennett; very 2021 book cover). I also finished a book in Romanian, translated from Russian: Zuleiha opens her eyes (Guzel Yakhina. Really beautiful, much more hopeful and uplifting than the Russian literature I'm used to.)], [After 15+ years of living in North America, I think I am "into" oatmeal. Oatmeal isn't a Romanian thing at all – oats are for horses, cream of wheat is for babies, adults don't eat either. Current approach: 1. overnight oats (in water, cashew milk makes them too... intense and I don't give money to the meat industry for cow milk) + either apple sauce (that I make because I keep forgetting to eat apples), or frozen fruit.], [Other food shit I'm obsessed with right now: ingesting industrial quantities of watermelon, these keto peanut butter cookies (I don't eat keto, but pb cookies with no-guilt-sugar are delicious), cashew milk.], [Arts: I am deeply struggling with a 3 layer linocut. I've carved it like 3+ times, and no matter how I do it I can't line up all the 3 layers well. I think it's time to give up

standard-article(title: [Episode 226: A Wild One], author: [Nate DiMeo], source-name: [The Memory Palace], images: (), paragraphs: ([Order The Memory Palace book now, dear listener. On Bookshop.org , on Amazon.com , on Barnes & Noble , or directly from Random House . Or order the audiobook at places like Libro.fm .], [The Memory Palace is a proud member of Radiotopia from PRX. Radiotopia is a collective of independently owned and operated podcasts that's a part of PRX, a not-for-profit public media company. If you'd like to directly support this show, you can make a donation at Radiotopia.fm/donate. I have recently launched a newsletter. You can subscribe to it at thememorypalacepodcast.substack.com .], [Order Eliza McGraw's wonderful new book, Astride: Women, Horses and a Partnership that Changed America.], [Music Hallelujah from Neu!, basically one of the best songs there is. Notes There is a lot written about Bessie, including some wonderful children's books. The best thing I read about her was in Ann Farrar (a journalist and biker herself)'s book Hear Me Roar: Women, Motorcycles, and the Rapture of the Road.],), insert-map: (:), word-count: 162, edited-for-length: false, debug-mode: false,)

standard-article(title: [What Do Social Media Companies Fear? Time Management.], author: [Study Hacks], source-name: [Study Hacks (Cal Newport)], images: (), paragraphs: ([I recently came across an interesting academic article in the journal Frontiers in Psychology. It was titled, "The relationships between social media use, time management, and decision-making styles."], [The paper's author surveyed 612 university students and young adults, asking them, among other things, about their digital habits and levels of personal organization. Using a linear regression analysis, she uncovered the following:], ["Social media use was negatively and significantly associated with overall time management and all its subscales."], [Here's the standard interpretation of this result: Social media is distracting, and if you're distracted, it becomes harder to maintain control over your schedule. So, the more you use social media, the worse you become at time management.], [But I've become interested in the reverse form of this argument: the better your planning system, the less time you'll spend on engagement-based applications like social media .], [Here's my thinking...], [When you're following an intentional schedule, your efforts are oriented toward goals that you find important. You also feel a satisfying sense of self-efficacy. These realities engage your long-term reward system, which can override the urges generated by its short-term counterpart, dissipating the drive for quick gratification from activities like glancing at your phone.], [In other words: The more you organize your analog life, the less appealing you'll find the digital alternative.], [If this is true, then maybe the thing social media companies

on it and make it into a digital print. Also, I've been working on L-systems, which make pretty trees. Should I write a blog post about that when it's done?],), insert-map: (:), word-count: 328, edited-for-length: false, debug-mode: false,)

fear most is not some newly-powerful application-blocking software or impossibly strict regulation, but rather a good old-fashioned daily planner.], [A lot of people I know have been freaked out recently by a viral essay with a grandiose title: “Something Big is Happening.” I recently released a short video in which I conduct a close analysis of this piece. (Spoiler alert: I wasn’t impressed.) Check it out.], [(More generally, I’ve been considering starting a separate weekly podcast/newsletter dedicated to providing a reality check on recent AI news. It feels like it might be useful to separate this discussion from my existing podcast and newsletter, which are more focused on how individuals can seek depth in a distracted world. But also, maybe this is a bad idea? I’m interested to hear your thoughts about this plan.)], [The post What Do Social Media Companies Fear? Time Management. appeared first on Cal Newport .],), insert-map: (:), word-count: 386, edited-for-length: false, debug-mode: false,)

So, the more you use social media, the worse you become at time management.

Study Hacks

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Episode 166: The Silent Room

The Memory Palace

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A note on shownotes. In a perfect world, you go into each episode of the Memory Palace knowing nothing about what’s coming. It’s pretentious, sure, but that’s the intention. So, if you don’t want any spoilers or anything, you can click play without reading ahead.

Dream by John Cage

The Introduction and Chi C’e Per Farmi i Rici from The Girl of the Golden West

The story of John Cage in the anechoic chamber was first told to me maybe twenty years ago by my friend Dave Panosky one night while walking around Providence. It was precisely the type of story that I wanted to one day make a show to tell. You can find it in a lot of places including in Cage’s own writing.

I first came across the second silent room in Alex Ross’ wonderful, The Rest is Noise .

I also tapped into Puccini and the Girl: History and Reception of Girl of the Golden West by Annie Randall and Rosalind Gray Davis.

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It’s pretentious, sure, but that’s the intention.

The Memory Palace

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ANALYSIS

Lighting an Ikea Billy Bookcase

Erik Runyon

This past December I had the opportunity to put together my first proper home office. One aspect of the office is a bookshelf to corral all my books and memorabilia that up to this point have been scattered around the house. For this project I utilized three Ikea Billy bookcases, but to make the project really shine, I added LED lighting. Let's go over the products and process I followed to set this up.

id="tldr">TL;DR

LED and wire channels = clean finish

Cable management = worth the effort

Splicing wires = most time-consuming

End result = cozy, nerd-approved display

id="shopping-list">Shopping List

Ikea Billy bookcase

Ikea HÖGBO glass doors

LED Strip Lights - 16.4ft 3000K Dimmable

LED Strip Channel Diffuser

Black cable channel

22/2 AWG Red/Black Electrical Wire

20 Pack COB 0.31in(8mm) 2-pin Transparent Connector

id="tools-needed">Tools needed

Electric drill for drilling wiring holes in the back of the case and attaching difuser channel

Hacksaw for cutting difuser channel

Tin-snips for cutting cable channel

Screwdriver and allen wrenches

Two of the three shelves assembled and ready for wiring

The HÖGBO doors come with inserts to hide the contents, but since the whole point here is to enjoy the display, I left those off. The reason for the doors is to keep everything inside relatively dust-free. These doors are made for the Billy cases, so they're relatively easy to install.

id="step-1-led-channel">Step 1: LED channel

LED diffuser channel installed on underside of shelf

A big priority for this project was to make the lighting look as clean as possible. So for step one I cut and installed the channel diffuser along the front lip of shelf. The diffuser

channel is alumnnium, so I used a hacksaw to cut to length. Luckily there was enough space between the shelf mounts and the front lip. To allow space for the wiring to run from the diffuser channel to the cable channel (see step #2), I cut a small notch.

Diffuser channel with notch for wiring

To connect the LED strip to the wiring I used 2-pin transparent connectors. This way the LED strip was only running along the front of the shelf, and the wiring could then do a 90-degree turn to head out the back of the case. If you look close, you would be able to see the connectors at the right end of the LED strip near the wall of the shelf. But I made sure these connections were at the door end of my office, so they're less noticeable.

id="step-2-cable-channel">Step 2: Cable channel

Diffuser and cable channel installed

The cable channel is added to the underside of each shelf to hide the wiring that runs from the LED strips to the back of the shelf where the wiring is joined. For this I drilled a hole just large enough for the wiring. This is a peel-and-stick product, so installation was simply a matter of cutting the plastic channel to fit, and then sticking it to the underside of the shelf. NOTE: Due to how I ran the wiring through the back of the shelf, it does not allow for future adjustment to the height of the shelves. It's not an issue in my case since I was going for a symmetrical, even-height look.

id="step-3-connecting-the-wires">Step 3: Connecting the wires

The LED kit I purchased came with two 5-meter (16.4 ft) LED strips. The included controller had two outputs, one for each strip. Since I'm lighting nine separate shelves and the top of the cases, this required quite a bit of wire splicing. But since all of the cables were exiting through the back of the cases, it was easy to hide the mess. Wiring was easily the most labor intensive part of the project as there was quite a bit of experimentation and testing before I found a solution that I liked and worked well.

One example of splicing the wires running to each strip

id="step-4-finish-adding-light-strips">Step 4: Finish adding light strips

All wiring connected

I forgot to take some intermittent shots at this juncture. I installed LED strips along the underside of each shelf for an even glow across all levels. Connecting the light strips was probably the easiest and fastest step in the process. As you can see in the image above, none of the red/black wiring is visible. All that's left is to finish attaching the top LED strip. After that comes the fun part... DECORATING!

id="the-finished-product">The finished product

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Hallucinating with art models

Monica Dinculescu

Wow, long time, no posts! Anyway, about them text-to-art generative models going about, eh? Surprising nobody: I am extremely into them. I've been using DALL-E and MidJourney since they came out, and even though tons has been written about them, I wanted to give a slightly different overview: the perspective of someone who isn't interested that much in their realism skills.

I think that the most compelling place for ML models in an artist's life is as a tool that specifically enables , and doesn't replace , creativity. Machine Learning is amazing at doing something very specific, lots of times, really fast. It's great at telling me if an image is a cat or a dog. It's also great at generating one hundred half-dog-half-cats, in different positions, so that I can bypass the dozens of hours I would spend sketching out half-dog-half-cats for a painting that's actually about the nuclear apocalypse.

I've seen a lot of examples of which model is best at painting "The otter with the pearl earring", but I haven't seen a lot of comparisons of these models in terms of their potential for creativity- likely because "creativity" is not really quantifiable. I wanted to do this for myself, if anything so that I can figure out how to use my money and credits better, but thought that I might as well put it out there in case anyone else was curious.

This post ended up being loooooong, so here's a Table of Contents:

Boring uses of interesting models

The (barely scientific) method

Cherry-picking outputs

What I look for

1. "Linocut print of a girl bundled up in bed with a stack of books and a cat"
2. "Lithograph of an orchid where each flower has a small skull inside"
3. "Erik Johansson photograph of a woman[sic] hair that is a literal bee hive"
4. "A toucan wearing a 60s apron, sitting on a mid century modern armchair, talking on a rotary phone, retrofuturism"

What have I learned?

id="boring-uses-of-interesting-models">Boring uses of interesting models

I use these new models for a very specific thing, and that is as a brainstorming partner/collaborator. This shouldn't

surprise you: I've been going off on using ML for co-creation since the days of working on Magenta , which was the project that forged my views on the topic (shoutout to the smarter-than-me people on the team that let me absorb their well articulated opinions).

I tend to find DALL-E generations like "dogs playing poker" or "Donald Trump but as a cheeseburger" impressive, but kind of boring: yes, the output is funny, but no, I don't fall asleep thinking about the deep meaning of an orange cheeseburger with a balding head and very small hands. As an artist who is trying to carve a place for myself in the art world (and for these models in my art), I want them to be my co-creators, my partners in crime, but not take over and compromise my style. I want us to brainstorm together, come up with ideas, and then (in most cases), mold this draft into something that I can look at and say "yeah, that looks like something I made".

This is consistent with how I look at the use of music models like the Music Transformer : absolutely impressive compositions, best suited for an elevator. This isn't the fault of the model, nor its users, and I am truly not shitting on these outputs (unless they're for NFTs; always here to shit on NFTs ☹): I think both background music and memes have a value in society, as do procedural TV shows (they're filming new Criminal Minds!!!), computer generated or not. But personally, as an artist, I feel fairly emotionally detached from them, as I'm pretty sure their authors do.

The exact same models, when used in a thoughtful and creative way lead to absolutely brilliant things; the ones that make you say "fuck, I wish I had thought of that". Yacht's album made with hard creative rules and machine learning is a wonderful example, as is Karen X. Cheng's use of DALL-E to imagine the scene beyond Girl with a Pearl Earring . This is the bit that I care about.

I've been trying to post updates about how I personally use these models on my artstagram (often with a commentary about the process), but as a recent craft fair that just rejected me had to say: I'm quite shit at the socials. Have this blog post instead?

id="the-barely-scientific-method">The (barely scientific) method

id="models">Models

I used the 4 models I have access to:

DALL-E , via the OpenAI website

Stable Diffusion , via the collab

MidJourney , via their Discord bot

DALL-E mini , via the HuggingFace interface.

I tried to get a Googler to help me run the prompts on Imagen , but I got no bites, so I have no idea how it fits into this story.

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Public Environment Variables For Your Tests Using cypress-expose Plugin

Gleb Bahmutov

Recently Cypress announced a change in how it will handle environment variables. Variables were always public and accessible to the application under test in the browser (via calling `Cypress.env()` method), but now you can have truly private values. I published a blog post [Migrating From Cypress.env To cy.env and Cypress.expose Methods](#) describing the change.

Looking at the changes, I noticed a missing area: it is not easy to pass public variables (non-secrets) to Cypress tests. I used a lot of “configuration” values that were meant to be used inside the spec files, and now these values are private by default. For example, let’s say we want to run a test only on CI. We could do something like this:

```
cypress/e2e/ci.cy.js 1 2 3 4 5 6 7 8 9 // only run these tests
on CI if ( Cypress . env ( 'ci' ) ) { it ( 'has CI variable' , () =>
{ expect ( Cypress . env ( 'ci' ), 'CI variable' ). to . equal ( true )
expect ( Cypress . env ( 'ciName' ), 'CI name variable' ). to .
equal ( 'GitHub Actions' , ) } }
```

In Cypress before v15 we could have set / used “ci” and “ciName” via environment variables, for example, if using Cypress github action :

```
.github/workflows/ci.yml 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
17 18 name: ci on: push
```

```
jobs: test: runs-on: ubuntu-24.04 steps:
```

- name: Checkout

```
# https://github.com/actions/checkout uses: actions/check-
out@v6
```

- name: Run tests

```
# https://github.com/cypress-io/github-action uses: cy-
press-io/github-action@v7 env: # default CI variable
CYPRESS_ci: true CYPRESS_ciName: 'GitHub Actions'
```

The test runs, `Cypress.env()` returns `{ ci: true, ciName: 'GitHub Actions' }`, everyone is happy. But in Cypress v16 the process environment variables that start with `CYPRESS_` are automatically added to the private `cy.env` space, and are not accessible outside test blocks / hooks. We could still pass them, but now we need to use a complete custom cypress `run -expose ...` command, instead of letting Cypress GitHub Action do its thing!

This is why I wrote cypress-expose plugin. It grabs every `CYPRESS_EXPOSE_...` process environment variable and sets it into `Cypress.expose` object, letting the specs access it. This is meant for public data only! Simply add plugin to your `setupNodeEvents` callback function and pass the config object.

```
cypress.config.js 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
19 const { defineConfig } = require ( 'cypress' ) const cypres-
sExpose = require ( 'cypress-expose' )
```

```
module . exports = defineConfig ( { allowCypressEnv : false ,
e2e : { // baseUrl, etc supportFile : false , fixturesFolder :
false ,
```

```
setupNodeEvents ( on, config ) { cypressExpose ( config)
```

```
// IMPORTANT: return the config object // to let Cypress
know we modified it return config } , }
```

Boom, you are all set. Now any `CYPRESS_EXPOSE_...` variable is available under camel-cased name. Let’s say we run Cypress using the following command:

```
1 2 3 $ CYPRESS_API_KEY=123secret! \
CYPRESS_EXPOSE_ANSWER=42 \ cypress open - env
PASS_WORD=my-password -expose USER_NAME=Joe
```

The api key and the password are going to be reachable via `cy.env` command, while the answer and the username are publicly accessible using `Cypress.expose` :

The `EXPOSE_ANSWER` value in the `cy.env` is null , the plugin overwrites it, but cannot delete it (since Cypress merged any modifications to the env object with the original values). I feel like having null there is enough.

Take the plugin for a spin, let me know if it works for you.

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Personal tech, social media, and the “decline of humanity”

Peter Dizikes | MIT News · MIT News Education

Social psychologist Jonathan Haidt presented a forceful analysis of the damage smartphones and social media are doing to our cognition, our civic fabric, and our children's wellbeing, while calling for renewed action to ward off their effects, in the latest of MIT's Compton Lectures on Wednesday.

“Around the world, people are getting diminished,” Haidt said. “Less intelligent, less happy, less competent. And it's happening very fast ... My argument is that if we continue with current trends as AI is coming in, it's going to accelerate. The decline of humanity is going to accelerate.”

Haidt is the Thomas Cooley Professor of Ethical Leadership at New York University's Stern School of Business and the author of the recent bestseller “The Anxious Generation,” which suggests that the widespread adoption of social media in the 2010s has been especially damaging to young women, making them prone to anxiety and depression.

But as Haidt has continued to examine the effects of social media on society, he has started focusing on additional issues. Our inability to put our phones away, our compulsion to check social media, and the way we spend hours a day watching short-form videos, may be causing problems that go far beyond any rise in anxiety and depression.

“It turns out, it's not the biggest thing,” Haidt said. “There's something bigger. It is the destruction of the human capacity to pay attention. Because this is affecting most people, including most adults. And if you imagine humanity with 10 to 50 percent of its attentional ability sucked out of it, there's not much left. We're not very capable of doing things if we can't focus or stay on a task for more than 30 seconds.”

Whatever solution may emerge to these problems, Haidt declared, is going to have to come from “human agency. People see a problem, they figure out a way around it. That's what I'm hoping to promote here [to] this very important audience. So please consider what I'm saying, these trends, and then work to change them.”

Haidt's lecture, titled, “Life After Babel: Democracy and Human Development in the Fractured, Lonely World That Technology Gave Us,” was delivered before a capacity audience of over 400 people in MIT's Huntington Hall (Room 10-250).

The lecture spanned a variety of related topics, with Haidt presenting chart after chart showing the onset of declines in cognition, educational achievement, and happiness, which all have seemed to occur soon after the widespread adoption of smartphones in the 2010s. The individual adoption of smartphones, he notes, has been compounded by the way schools brought internet-connected computing devices into classrooms around the same time.

“The biggest, the most costly mistake we've ever made in the history of American education [was] to put computers and high tech on people's desks,” Haidt said.

Distractible students with shorter attention spans are reading fewer books, he noted; some cinema students cannot sit through films. The top quartile of students is continuing to do well, he noted, but for most students, proficiency levels have dipped notably since the 2010s.

“Fifty years of progress in education, 50 years of progress, up in smoke, gone,” Haidt said. “We're back to where we were 50 years ago. That's pretty big, that's pretty serious.”

As Haidt mentioned multiple times in his remarks, he is not an opponent of all forms of technology, or even personal communication technology, but rather is seeking to mitigate its harmful effects.

“I love tech, I love modernity, we're all dependent on it, I love my iPhone,” Haidt said. Just as he finished that sentence, an audience member's cellphone started ringing loudly — drawing a huge laugh from the audience.

“I did not plant that, that was a truly spontaneous demonstration of what I'm talking about,” Haidt said.

Haidt was introduced by MIT President Sally A. Kornbluth, who called him “a leading voice for reforming society's relationship with technology.” She praised Haidt's work, noting that he wants to “encourage us to imagine a more positive role for technology in humanity's future.”

The Karl Taylor Compton Lecture Series was introduced in 1957. It is named for MIT's ninth president, who led the Institute from 1930 to 1948 and also served as chair of the MIT Corporation from 1948 to 1954.

Compton, as Kornbluth observed, helped MIT evolve from being more strictly an engineering school into “a great global university” with “a new focus on fundamental scientific research.” During World War II, she added, Compton “helped invent the longstanding partnership between the federal government and America's research universities.”

Haidt received his undergraduate degree from Yale University and his PhD from the University of Pennsylvania. He taught on the faculty at the University of Virginia for 16 years before joining New York University. He has written several widely discussed books about contemporary civic life. Haidt observed that the problems stemming from device distraction and compulsion appear to have hit so-called Gen Z — those born from roughly the mid 1990s to the early 2010s — especially hard, though he emphasized that people in that cohort are essentially victims of circumstance.

“I am not blaming Gen Z,” Haidt said. “I am saying we raised our kids in a way — we allowed the technology companies to take over childhood. We allowed a few giant companies to own our children's attention, to show them millions of short videos, to destroy their ability to pay attention, to stop

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standard-article(title: [Episode 231: On Dexter Avenue], author: [Nate DiMeo], source-name: [The Memory Palace], images: (), paragraphs: ([Order The Memory Palace book now, dear listener. On Bookshop.org , on Amazon.com , on Barnes & Noble , or directly from Random House . Or order the audiobook at places like Libro.fm .], [During mid-May, 2025, I'm doing a Midwestern book tour, with stops in Minneapolis, Cincinnati, Indianapolis, and Chicago. Find out more at www.thememorypalace.us/events .], [The Memory Palace is a proud member of Radiotopia from PRX. Radiotopia is a collective of independently owned and operated podcasts that's a part of PRX, a not-for-profit public media company. If you'd like to directly support this show, you can make a donation at [Radiotopia.fm/donate](https://radiotopia.fm/donate). I have recently launched a newsletter. You can subscribe to it at thememorypalacepodcast.substack.com .], [You can access the self-produced history of the Sophie Bibb Chapter of the UDC here. The Alabama Encyclopedia site does a nice job with some of this stuff. You might want to check out Caroline Janney's book, *Burying the Dead but Not the Past: Ladies' Memorial Associations and the Lost Cause.*],), insert-map: (:), word-count: 183, edited-for-length: false, debug-mode: false,)

standard-article(title: [From The Memory Palace Audio-book: Fine New England Granite], author: [Nate DiMeo], source-name: [The Memory Palace], images: (), paragraphs: ([Order The Memory Palace book now, dear listener. On Bookshop.org , on Amazon.com , on Barnes & Noble , or directly from Random House .], [Buy the audiobook wherever you get audiobooks (like libro.fm!)], [The Memory Palace is a proud member of Radiotopia from PRX. Radiotopia is a collective of independently owned and operated podcasts that's a part of PRX, a not-for-profit public media company. If you'd like to directly support this show, you can make a donation at [Radiotopia.fm/donate](https://radiotopia.fm/donate). I have recently launched a newsletter. You can subscribe to it at thememorypalacepodcast.substack.com .], [Audio excerpted courtesy of Random House Audio from THE MEMORY PALACE by Nate DiMeo, read by a full cast. Excerpt read by Nate DiMeo, © 2024 Nate DiMeo, © 2024 Penguin Random House, LLC. All rights reserved.],), insert-map: (:), word-count: 132, edited-for-length: false, debug-mode: false,)

standard-article(title: [I made a 2001-era emoji font! That you can use!], author: [Monica Dinculescu], source-name: [Monica Dinculescu], images: (), paragraphs: ([You know the scenes in Friends when Ross starts talking about dinosaurs and he's SUPER excited but everyone else is losing the will to live? This is basically that, only instead of dinosaurs, it's emoji, and unlike Ross, I have never successfully befriended a monkey.], [Last month, my coworker casually told me he still has a 2001 era DoCoMo phone, which is one of the first phones to have emoji (👉👈: emoji first appeared in 1999, on DoCoMo phones, and DoCoMo phones alone). So I got ahold of this phone. Which

standard-article(title: [Episode 158: Life's Work], author: [The Memory Palace], source-name: [The Memory Palace], images: (), paragraphs: ([Help support this show and the network that makes it possible by making a donation today. Make your mark at [Radiotopia.fm/donate](https://radiotopia.fm/donate)], [A note on shownotes. In a perfect world, you go into each episode of the Memory Palace knowing nothing about what's coming. It's pretentious, sure, but that's the intention. So, if you don't want any spoilers or anything, you can click play without reading ahead.], [I strongly recommend checking out Annelise Orlick's *Common Sense and a Little Fire*, an exhilarating history of Pauline Newman, Rose Schneiderman and other women at the center of the labor movement in the 20th Century.], [Learn about your ad choices: dovetail.prx.org/ad-choices],), insert-map: (:), word-count: 128, edited-for-length: false, debug-mode: false,)

standard-article(title: [Episode 92 (Oil, Water)], author: [The Memory Palace], source-name: [The Memory Palace], images: (), paragraphs: ([The Memory Palace is a proud member of Radiotopia, from PRX, a curated network of extraordinary, story-driven shows. Learn more at radiotopia.fm], [SPOILERS BELOW], [Notes *Great stuff in the Cleveland Plain Dealer from waaaaaay back to fire #1, if you want to dive in. * Found Jonathan Joseph Wlasiuk's dissertation , *Refining Nature* (etc.) quite helpful in sorting out the early days of the Rockefeller refineries. * If you want to know more about the complicated relationship of Cleveland and the '69 fire and the passage of the Clean Water Act, seek out Jonathan H. Adler's article (and R. E. M. nod), *Fables of the Cuyahoga: Reconstructing a History of Environmental Protection* . * I also want to shout out *The Killer in the Attic*, and *More True Tales of Crime and Disaster from Cleveland's Past*, by John Stark Bellamy II, which does a great job with the river fires.], [Music * Start off with *Lacrymae* , from *Melodium*. * Go to a chopped up *Fables* , by *Girls in Airports*. * Finish off with the eternal *Sunflower River Blues* by John Fahey.], [Learn about your ad choices: dovetail.prx.org/ad-choices],), insert-map: (:), word-count: 187, edited-for-length: false, debug-mode: false,)

standard-article(title: [Seminoles Taught American Soldiers a Thing or Two About Guerrilla Warfare], author: [Jon Guttman], source-name: [HistoryNet], images: (), paragraphs: ([The word "Seminole" is derived from the Muscogean word *simanó-li* , or "runaway," reflecting a common heritage, as Upper Creeks from Alabama, Lower Creeks from Georgia, other affiliated tribes and escaped African slaves all sought sanctuary in Spanish Florida. There they mixed with one another, adapted to their surroundings, traded with Britain, Spain and the United States and came to be collectively recognized as one of the Five Civilized Tribes (along with the Cherokee, Chickasaw, Choctaw and

charged, turned on and most importantly, TOTALLY had OG emoji:], [style="width: 100%;">], [I spent the whole day being unproductive and sending emoji messages to people:], [style="width: 100%;">], [id="svgs">SVGs], [I then took a 10 hour flight to Europe and, for lack of better things to do while watching every movie that came out this year, I drew every one of those emoji as a sprite. 166 emoji in total, 12x12px each, in one of six colors. This was my first time doing pixels sprites, so I obviously fucked it up: I ended up with a bunch of random sprite sheets, each with a random number of sprites in it, which was a bit of a mess. Thankfully, Amanda Glosson , reigning queen of pixels, wrote me a script to transmogrify my mess into individual svgs. These individual SVGs, to be exact:], [LOOK HOW PRETTY THEY ARE! 🐊], [id="a-wild-font-appears">A wild font appears], [The reason why I made those SVGs was partly because Captain America: Civil War is unbearably boring, but partly because I wanted to make a font and use it everywhere like an emoji hipster.], [So then I did. I used Fontastic to make the font – it's black and white, because at the time of writing this, colour fonts weren't supported on the web (tbh, even a year later, when colour fonts are supported, I still couldn't tell you how to make on). I also mapped the original emoji glyph to one of the current existing emoji code points, based on this list, because let's be honest, some of them were mysterious. Do you know what 🐊 means? It's 🐊. And 🐊 is 🐊. 12 pixels ain't a lot of pixels, friends.], [Here the 166 emoji as they look today:], [And here's the same list, using the DoCoMo emoji font:], [There's some OG emoji that don't even exist today!:], [Boom! 🐊], [If you want to use it, you can download the font here , and use it as a font-face :], [If you're going to use it, maybe give me some credit, because I spent an unhealthy amount of time on it. Also, if you're DoCoMo, please don't sue me. Emojineering comes only from the ♥️.],), insert-map: (:), word-count: 487, edited-for-length: false, debug-mode: false,)

standard-article(title: [That Efficiency of Programming Languages Image], author: [astorm], source-name: [Alan Storm], images: (), paragraphs: ([In recent years there's this image that will make the rounds with a ranking of programming languages by energy efficiency. If you haven't seen it, it looks like this. I want to be careful here because I'm not a researcher. I don't design studies and I don't know how to design studies or anything about the environment folks will typically design studies in. This is intended as a critique of the methodology not a canceling of the research — if I'm off base here please let me know. The first thing that stood out to me was JavaScript had a [...]],), insert-map: (:), word-count: 101, edited-for-length: true, debug-mode: false,)

Creek/Muscogee) of the American Southeast. Not civilized enough for some, apparently, for when the United States acquired Florida in 1821, it began herding the Seminoles, through a succession of treaties, to progressively smaller and less desirable parts of the state. Ultimately, in 1834 federal officials set Jan. 1, 1836, as a deadline for removal of the Seminoles from Florida to Indian Territory (present-day Oklahoma). When a 110-man detachment of the 4th U. S. Infantry under Brevet Maj. Francis L. Dade set out for Fort King to oversee that final removal, however, it was ambushed by 180 Seminoles led by Halpatter Tustenuggee (Alligator) on Dec. 28, 1835. The ensuing slaughter of all but two of Dade's men marked the beginning of the longest campaign fought between the U. S. Army and American Indians, a mix of pitched battles and guerrilla warfare.], [In Seminole Warrior versus U. S. Soldier: Second Seminole War, 1835– 42 (No. 61 in Osprey's Combat series) military historian Ron Field compares the tactics, arms, equipment and fighting techniques used by both sides. In addition to what whites termed the "Dade massacre," re-examined here through a more impartial lens, Field relates two other major engagements—at Lake Okeechobee, on Christmas Day 1837, and the second of back-to-back clashes at the Loxahatchee River, on Jan. 24, 1838.], [Set in what might be more appropriately called the "Wild South," the Second Seminole War was something of a learning process for both sides. Several notable future U. S. Army figures had their first taste of Indian warfare and, for that matter, their first combat experience during the conflict, lessons they would apply during the Mexican War. The Seminoles likewise developed guerrilla tactics they would employ as scouts during the Army's later run-ins with other tribes on the Great Plains. Backed by Osprey's usual fine array of illustrations and maps, Seminole Warrior versus U. S. Soldier offers Wild West readers a peek at the dress rehearsal for Indian campaigns to come.], [Second Seminole War, 1835–42], [By Ron Field, Osprey Publishing, 2022], [If you buy something through our site, we might earn a commission.],), insert-map: (:), word-count: 443, edited-for-length: false, debug-mode: false,)

standard-article(title: [TMR 24.9: "I Hate Thinking about Thinking about Myself Like This" [Melvill]], author: [Chad Post], source-name: [Three Percent (U of Rochester)], images: (), paragraphs: ([In the penultimate episode of this season, there's a lot of talk about Melville's relationship with Nathaniel Hawthorne as we get more of Melville's life as he bounces through time, writes Moby-Dick and the creation of "its own kind of reader: an inexhaustible reader. A reader that didn't yet exist . . ."], [The " Grifters Gonna Grift " t-shirt is still available and still sexy.], [Next episode will be next week at the regularly scheduled time and will cover the end of Rodrigo Fresán's Melvill . You can find the full reading schedule here .], [This week's music is " Vertigo " by FM Belfast.], [You can find all previous seasons of TMR on our YouTube channel and on Apple Podcasts , Spotify , etc. Please rate and review! It helps more than you know.], [Follow Open Letter, Two Month Review , Chad Post, Kaija Straumanis , and Brian Wood for random thoughts and information about upcoming guests.],), insert-map: (:), word-count: 159, edited-for-length: false, debug-mode: false,)

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BRIEFS

IN BRIEF

New Scientist, New Scientist: In a randomised trial, men who experience premature ejaculation benefitted from using an app to learn techniques for extending intercourse

The Memory Palace, The Memory Palace: If you enjoy this story, do tell someone about The Memory Palace.

Thanks.

Nate

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Music

We hear both Baracolle and Ice World from Saloli's album, The Deep End .

We hear Ensemble Recherche do Morton Feldman's Something Wild in the City: Mary Ann's Theme.

As well as De Wolfe Music do Moonbird .

Harry Kalahiki does that Ukelele version of Claire de Lune

We get a snippet of Eartheater's Peripheral

And E Ruscha V's The Hostess

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Music

Traffico from Carlo Rustichelli's score to Divorzio All'Italiana and Una Braveta, from his score to Amici Miei. Tema Grottesco from Giovanni Fusco's score to L'avventura. Gloving it from Moondog. Musica Bionda from the score to The Sweet Body of Deborah The Peter Thomas Sound Orchestra plays Natascha from the score to The Perfect Marriage. Waltz from the Brendan Eder Ensemble Christa Schonfeldinger plays Grieg's Smarthold - der Kobold, on the glass armonica.

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A note on notes: We'd much rather you just went into each episode of The Memory Palace cold. And just let the story take you where it well. So, we don't suggest looking into the show notes first.

Music

Midnight Moon from Annalisa Tornfelt and Gideon Fraudmann's album, Spring Breakup: Songs for Alaska.

A couple of numbers from Marcelo Zarvos' wonderful score to the wonderful, Please Give.

A remarkable piece of music called Spectacle of Ritual by Kali Malone

And another called Pauvre Simon from Sylvain Chauveau

New Scientist, New Scientist: This is the opening of Kim Stanley Robinson's Red Mars, the New Scientist Book Club read for April, as humans come to the planet to settle it

New Scientist, New Scientist: We know that a person's outlook can have a huge effect on their health, and it's no different when it comes to ageing. Columnist Graham Lawton looks at new evidence of just how powerful our attitude is – and how to use it to age better

New Scientist, New Scientist: Ancient DNA reveals that the Goths of eastern Europe, some of whom would ultimately sack the city of Rome, may have been a mix of peoples from three continents

New Scientist, New Scientist: Toys powered by AI show a worrying lack of emotional understanding. But we need to understand the risks and benefits of the technology so the industry can be regulated, not outright banned

Nate DiMeo, The Memory Palace: Order The Memory Palace book now, dear listener. On Bookshop.org , on Amazon.com , on Barnes & Noble , or directly from Random House . Or order the audiobook at places like Libro.fm .

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Music

On-ness by Tom Rogerson and Eno. Etude by Joep Beving Ebb Tide by Houston & Dorsey

jessica.blake@insidehighered.com, Inside Higher Ed: New Survey Announcements From IES Give Hope for Agency's Revival

class="field field--name-field-featured-image field--type-entity-reference field--label-hidden field__item">

The Memory Palace, The Memory Palace: If you enjoy this story, do tell someone about The Memory Palace. Living parts by killing a bacterial cell and then transplanting the genome of another species into it, blurring the boundary between life and death

Thank you kindly: 3:00 AM

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